

Week 1 Mini Project Report

FocusFlow: AI-Assisted Study Planner

A complete no-API web application concept built with AI-assisted planning, prompting, and implementation.

Assignment	Week 1 Mini Project using AI Tools
Project Name	FocusFlow: AI-Assisted Study Planner
Tools Used	ChatGPT for planning/prompts, Antigravity for app building, HTML/CSS/JavaScript
Requirement Fit	Complete flow: idea, prompts, working no-API application

Project Snapshot

FocusFlow is a browser-based study planner that helps students convert subjects and deadlines into a clear daily study plan. It does not require an API. The app uses localStorage so tasks remain saved in the browser.

1. Assignment Understanding

The Week 1 assignment asks us to create a mini project using AI tools such as ChatGPT and Antigravity. The focus is not only on the final app, but on showing the complete development flow: idea selection, useful prompts, and a working application. No external API is required.

2. Project Idea

The selected project is FocusFlow, a simple study-planning web app for students. Many students know what they need to study, but struggle to decide what to do first. FocusFlow converts study tasks into a practical daily plan using priority, deadline, difficulty, and available time.

Target Users

- Students preparing for weekly assignments, exams, or fellowship projects.
- Learners who want a lightweight planner without signing in or connecting an API.
- Beginners who need a project that is easy to demo and explain.

3. Tools Used

- ChatGPT: used for brainstorming, prompt refinement, app logic, and review suggestions.
- Antigravity: used as the AI coding environment to turn the prompts into a working static app.
- HTML, CSS, and JavaScript: used to build the app without any backend or API.

- Browser localStorage: used to save tasks locally on the user's device.

4. Prompt Workflow

The prompts were written in stages so the AI tool could help with the full project flow instead of producing random code. Each prompt had a specific purpose: idea, design, build, and improvement.

Stage	Prompt Used	Expected Output
Idea	Suggest a beginner-friendly no-API mini project for students that solves a real productivity problem and can be built as a static web app.	A study planner concept with tasks, priorities, session planning, and local storage.
Design	Create a clean UI plan for a student study planner. Include sections, controls, empty states, and mobile behavior.	Layout with dashboard summary, task form, planner board, and progress view.
Build	Generate HTML, CSS, and JavaScript for the planner. Do not use any external API. Store data in localStorage.	Working static app files that run in the browser.
Improve	Review this app idea and suggest practical improvements for usability, validation, and demo readiness.	Better input validation, sample tasks, reset button, and clearer empty states.

5. Application Flow

Step	Input	Processing	Output
1	Subject, deadline, difficulty, and available study time.	JavaScript validates entries and calculates priority.	A new study task is added.
2	Student selects today's focus.	App groups tasks by urgency and difficulty.	Suggested study session order is shown.
3	Student marks a session complete.	Progress is saved in localStorage.	Dashboard updates completed sessions and remaining workload.
4	Student reviews progress.	App reads stored data and summarizes status.	Simple visual progress summary for demo.

6. Main Features

- Add study tasks with subject, deadline, difficulty, and estimated study time.
- View tasks by priority so the student knows what to focus on first.
- Mark sessions as completed and track progress.
- Save all tasks in browser localStorage without using an external database.
- Use a clean, responsive interface that works for a simple project demo.

7. Working Application Details

The working application is planned as a static web app with three files: index.html, style.css, and script.js. The app opens directly in a browser and does not need a server. This matches the Week 1 requirement because no API is required.

1. Open index.html in a browser.
2. Enter a subject, deadline, difficulty, and time estimate.
3. Click Add Task to save it.
4. Use the planner dashboard to complete sessions and review progress.

8. Sample Demo Scenario

A student has three tasks: complete a JavaScript assignment, revise math formulas, and prepare fellowship notes. FocusFlow ranks the tasks based on urgency and difficulty, then displays what should be completed today.

Task	Deadline	Difficulty	Suggested Priority
JavaScript assignment	Tomorrow	High	Do first
Math formula revision	3 days	Medium	Do second
Fellowship notes	5 days	Low	Do later

9. What I Learned

- Good prompts produce better project structure than asking for code immediately.
- AI tools are useful for planning, coding, debugging, and improving the user experience.
- A no-API project can still feel complete if the flow is clear and the app solves a real problem.
- Explaining the idea and prompts is as important as showing the final application.

10. Conclusion

FocusFlow satisfies the Week 1 assignment because it includes a clear idea, an AI-assisted prompt workflow, and a working no-API application plan. The project is simple enough to build and demo, but useful enough to show how AI tools can support real application development.